

The synergistic gelation of novel *Bletilla striata* polysaccharide with hyaluronic acid: Characterization, rheology

Didi Ma^{a,1}, Zhenzhen Zhao^{a,1}, Yujing Wen^{a,1}, Juan Zhou^a, Wenhao Zhou^a, Jian Mao^c, Kun Lv^{a,b,**}, Yunpeng Cao^{d,*}, Lan Jiang^{a,b,**}

^a Anhui Province Key Laboratory of Non-coding RNA Basic and Clinical Transformation (Wannan Medical College), Yijishan Hospital of Wannan Medical College, Wuhu 241000, Anhui, China

^b Central Laboratory, Yijishan Hospital of Wannan Medical College, Wuhu 241000, Anhui, China

^c Yangtze River Delta Information Intelligence Innovation Research Institute, Wuhu 241000, Anhui, China

^d CAS Key Laboratory of Plant Germplasm Enhancement and Specialty Agriculture, Wuhan Botanical Garden, Chinese Academy of Sciences, Wuhan 430074, Hubei, China

ARTICLE INFO

Keywords:

Bletilla striata
Polysaccharide
Structure
Rheology
Synergistic gelation

ABSTRACT

Bletilla striata polysaccharide (BSP) has attracted considerable interest due to its diverse biological activities. In this research, a novel low-molecular-weight water-soluble polysaccharide (BSP-182) was isolated and purified from *Bletilla striata* tubers, and its structure was characterized. The findings indicated that BSP-182 is predominantly composed of β -1,4-linked glucose (Glc) and β -1,4-linked mannose (Man) in a molar ratio of approximately 7.8:2.2. Additionally, hyaluronic acid (HA)/BSP-182 hydrogels were synthesized, and their physicochemical properties and structure were examined. Rheological analysis revealed that HA and BSP-182 form hydrogels via hydrogen bonding, with concentration-dependent enhancements in rheological properties, textural attributes, and thermal stability. The hydrogels displayed significant shear-thinning behavior and viscoelasticity, which are beneficial for food processing and texture modification, especially in the development of easy-to-swallow foods. This research offers valuable insights for the development of innovative BSP-based hydrogels for dysphagia management.

1. Introduction

Bletilla striata, with a usage history spanning thousands of years, is a significant traditional Chinese medicinal resource recognized in both the “Chinese Pharmacopoeia (2020 Edition)” and “Shennong’s Classic of Materia Medica” (He et al., 2017). It is a common component in many traditional Chinese herbal formulations, known for its diverse biological activities (Delin et al., 2019). We have revealed for the first time in a previous study the genome of a chromosome-level haplotype of *Bletilla striata* polysaccharide (BSP), analyzed the biosynthetic pathway of BSP

(Jiang et al., 2022). Recent studies have identified polysaccharides as the most crucial bioactive elements in the tubers of *Bletilla striata*. These polysaccharides demonstrate a range of biological activities, including immune regulation, anti-inflammatory, anti-tumor, anti-fibrotic, hemostatic, and gastroprotective effects (He et al., 2024; Li et al., 2023; Liu et al., 2022; Niu et al., 2022; Tang et al., 2024). Natural hydrogels, especially composite polysaccharide hydrogels derived from biopolymers such as polysaccharides, are attracting increasing attention. BSP, a promising natural functional polysaccharide with a molecular weight ranging from 100 to 300 kDa, comprises mannose and glucose

Abbreviations: BSP, *Bletilla striata* polysaccharide; HA, hyaluronic acid; CS, chitosan; TFA, trifluoroacetic acid; NMR, Nuclear magnetic resonance; COSY, correlated spectroscopy; NOESY, nuclear overhauser effect spectroscopy; HSQC, heteronuclear single quantum coherence spectroscopy; HMBC, heteronuclear multiple bond coherence spectroscopy; LVR, linear viscoelastic region; SEM, Scanning electron microscopy; FTIR, Fourier Transform infrared spectroscopy; HPLC, high-performance liquid chromatography; Mp, Peak Molecular Weight; Mw, weight-average molecular weight; Mn, number-average molecular weight; RT, Retention Time; Man, mannose; Glc, glucose; GC-MS, gas chromatography–mass spectrometry; PMAAs, partially methylated alditol acetates; DEPT, distortionless enhancement by polarization transfer; TG, thermogravimetric; DTG, derivative thermogravimetric; DSC, differential scanning calorimetric; G, storage modulus; G', loss modulus..

* Corresponding author.

** Corresponding authors.

E-mail addresses: lvkun@yjsy.com (K. Lv), xfcpeng@126.com (Y. Cao), jianglanhi@163.com (L. Jiang).

¹ These authors contributed equally to this work.

<https://doi.org/10.1016/j.foodchem.2024.142359>

Received 31 July 2024; Received in revised form 27 November 2024; Accepted 2 December 2024

Available online 5 December 2024

0308-8146/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

units linked by β -glycosidic bonds (Chen et al., 2018). As a water-soluble plant β -glucan, BSP exhibits excellent hydrophilicity, swelling capacity, and permeability. The abundance of hydroxyl groups within its structure provides numerous sites for chemical modification. Moreover, the inherent stability, biocompatibility, and mucoadhesivity of BSP render it a promising candidate for hydrogel fabrication (Sun et al., 2022; Yang et al., 2020). BSP-based hydrogels have been used for oral administration, including BSP/chitosan (CS) composite hydrogels (Wang et al., 2023) and BSP/sodium alginate in situ gels (Fan et al., 2024).

Composite polysaccharides have attracted significant attention in the food industry due to their unique gelling properties, overcoming the limitations of individual polysaccharides. Research has demonstrated that intermolecular interactions between polysaccharides significantly influence key performance characteristics of mixed gels, including gelling capacity, strength, and water-holding capacity (Luo et al., 2018; Zhao et al., 2023). These gel properties are crucial in food processing, contributing to improved food texture, moisture retention, and enhanced stability, ultimately facilitating easier swallowing (Bao et al., 2019). For instance, the feasibility of using inulin/konjac glucomannan (KGM) mixtures as a gelling matrix for dysphagia-oriented foods (Chen et al., 2024). A food-grade gel based on a gelatin-konjac glucomannan binary hydrogel system to achieve targeted oral sensations and modulate digestive behavior (Tian et al., 2024). Hyaluronic acid (HA) is a natural acidic mucopolysaccharide composed of a disaccharide repeating sequence of *N*-acetylglucosamine and glucuronic acid, linked together by β -1,4 and β -1,3 glycosidic bonds (Liu et al., 2011). HA has excellent moisturizing, lubricating and antioxidant properties, with a wide range of applications in pharmaceuticals, food, cosmetics, biomaterials and other fields (Avadhani et al., 2017; Liuying et al., 2022; Ni et al., 2023; Ren et al., 2022). HA also exhibits high hydrophilicity due to the abundant carboxylate (COO⁻) and hydroxyl (OH⁻) groups within its disaccharide units (Suh et al., 2005). This inherent hydrophilicity confers unique advantages in hydrogel formation, positioning HA as a promising novel food ingredient. Moreover, a multitude of HA-based oral hydrogels have been developed, including HA/chitosan (Lin et al., 2023), HA/whey protein isolate hydrogel (Wang et al., 2023), and HA/*Ocimum basilicum* seed mucilage/thiolated alginate hydrogel (Hosseini et al., 2022).

BSP and HA hydrogels offer potentially synergistic benefits as delivery vehicles. Both are biodegradable and endogenous, making them promising candidates for use as delivery vehicles. These inherent properties, coupled with the physical structure of the resulting gel, further enhance their potential in biomaterial applications (Cui et al., 2017; Huang et al., 2019). This physical crosslinking method, applicable to food-grade materials, enhances the mechanical properties of hydrogels under mild conditions, while avoiding the potential toxicity of certain chemical crosslinkers or initiators (Liu et al., 2022; Ma et al., 2021; Qiu et al., 2024). These properties indicate the potential of the composite hydrogel for developing swallowable foods enriched with the nutritional benefits of medicinal plants. Although numerous hypotheses have been proposed, the underlying mechanism of action of HA/BSP hydrogels remains elusive. This synergistic effect is likely influenced by a multitude of factors, including ionic strength, pH, and chemical structure (Wang et al., 2018). Based on the distinct physicochemical properties of HA and BSP, we hypothesize that a composite gel can be formulated. This novel gel holds promise as an effective delivery vehicle for dysphagia-friendly foods.

In this study, we initially isolated BSP via water extraction and ethanol precipitation. Subsequently, the structure of BSP was elucidated through a series of analyses including molecular weight determination, monosaccharide composition analysis, methylation analysis, and nuclear magnetic resonance (NMR) spectroscopy. Additionally, we examined the physicochemical properties and structural characteristics of hydrogels with varying BSP:HA ratios. This was achieved using rheology, infrared spectroscopy, and electron microscopy to shed light on the interaction mechanism between BSP and HA. This study offers a

theoretical foundation for the development and application of easy-to-swallow hydrogel-based foods.

2. Materials and methods

2.1. Materials and reagents

Dried *Bletilla striata* tubers were collected from Bozhou, Anhui, China. HA (purity $\geq 93\%$, food grade, molecular weight 200–400 kDa) were purchased from Macklin Chemical Co. (Shanghai, China). All other chemical reagents and solvents were of analytical grade unless specified.

2.2. Extraction and purification of BSP

Grind the dried *Bletilla striata* tuber into powder, pass through a 10-mesh, 2-mm pore size sieve to obtain *Bletilla striata* powder. Extract the *Bletilla striata* powder (200 g) with deionized water at a ratio of 1:10 at 90 °C for 2–3 times (Zhou et al., 2022). The concentrated extract was precipitated with four volumes of anhydrous ethanol at 4 °C for 12 h. The precipitate was collected by centrifugation at 5000 rpm for 10 min. Dissolve the precipitate in deionized water and perform protein depigmentation using the Seavage method (Yuan et al., 2020). Dialyze the protein-depigmented samples in a dialysis bag (1kDa) at 4 °C for 72 h. Collect the undialyzed solution and freeze-dry to obtain crude polysaccharides (Cao et al., 2024). The purification process utilized DEAE-52 cellulose (Solarbio, Beijing, China) and a fully automated gel purification system (BRT-GS, Borui Saccharide, Yangzhou, China). This system was integrated with online detection and collection capabilities, using a differential detector (Shimadzu) to isolate symmetrical peaks, thereby yielding a high-purity BSP.

2.3. Structural characterization of BSP

2.3.1. Monosaccharide composition

Transfer 5 mg of BSP into an ampoule containing 2 mL of trifluoroacetic acid (TFA, 3 M) and hydrolyze the mixture at 120 °C for 3 h (Synytsya et al., 2015). Extract the acid hydrolysis solution into a tube and dry it under nitrogen. Add 5 mL of water and vortex to mix thoroughly. Remove 50 μ L of this solution, add it to 950 μ L of deionized water, and centrifuge at 12,000 rpm for 5 min to separate the supernatant. Analyze the collected supernatant using an ion chromatograph (ICS5000, USA) equipped with a Dionex™ CarboPac™ PA20 chromatography column (3*150 mm) and an electrochemical detector (Ye et al., 2021).

2.3.2. Molecular weight determination

The samples and standards were accurately weighed, and the samples were prepared in a 5 mg/mL solution. After centrifugation at 12,000 rpm for 10 min, the supernatant was filtered through a 0.22 μ m microporous membrane. The filtered samples were then transferred to 1.8 mL vials for injection and characterized using a high-performance liquid chromatography system (LC-10 A, Shimadzu) (Tan et al., 2023).

2.3.3. Methylation analysis and nuclear magnetic resonance (NMR)

Dissolve 3 mg of BSP in DMSO and proceed to methylate the mixture using CH₃I in a DMSO and sodium hydroxide solution. After the methylation of the polysaccharide, add 2 M trifluoroacetic acid (TFA) and hydrolyze for 90 min. Subsequently, evaporate the mixture using a rotary evaporator. Reduce the residue with sodium borohydride for 8 h, and then acetylate with acetic anhydride at 100 °C for 1 h (Peng et al., 2024). Finally, dissolve the acetylated product in CH₂Cl₂ and analyze using an Agilent GCMS 6890–5973 gas chromatography–mass spectrometry system.

Dissolve 50 mg of BSP in D₂O and then subject it to freeze-drying. Acquire 1D (¹H and ¹³C NMR) and 2D NMR spectra—including correlated spectroscopy (COSY), nuclear overhauser effect spectroscopy

(NOESY), heteronuclear single quantum coherence spectroscopy (HSQC), and heteronuclear multiple bond coherence spectroscopy (HMBC)—of BSP at 25 °C using an NMR spectrometer (Bruker, Germany). Record the relevant data accordingly (Mzoughi et al., 2018).

2.4. Synergistic gelation of BSP and HA

2.4.1. Gel preparation

Dissolve BSP and HA (purity ≥ 93 %, molecular weight 200–400 kDa, sourced from Shanghai, China) powders in deionized water while magnetically stirring at 25 °C for 10 min to prepare fully dissolved solutions of BSP (0.5 %, w/v) and HA (1 %, w/v).

At 25 °C, where BSP-182 and HA were mixed and allowed to stand at this temperature for complete swelling and gel formation. BSP-182 molecules contain glucose and mannose, which are rich in hydroxyl groups capable of forming hydrogen bonds within the BSP-182 solution. HA, due to the abundant carboxylate (COO^-) and hydroxyl (OH^-) groups in its disaccharide units, exhibits high hydrophilicity and possesses unique advantages in hydrogel formation. These functional groups enable HA to readily form hydrogen bonds or van der Waals forces in aqueous solutions, leading to strong hydrogen bond interactions with other macromolecules. At room temperature (25 °C), the two components form a hydrogel through covalent chemical and physical crosslinking mediated by hydrogen bonding and electrostatic interactions. The formation of hydrogen bonds in the hydrogel is not a process that occurs at a specific temperature. Rather, it is a dynamic process that is influenced by temperature and other factors.

Subsequently, mix the BSP and HA solutions uniformly under magnetic stirring at 25 °C for 10 min to achieve different mixed ratios (HA at 7:3, 5:5, and 3:7). Allow the mixtures to sit at room temperature (25 °C) for sufficient swelling and gel formation (Hou et al., 2023b). Prior to testing, let all samples equilibrate at room temperature for 12 h.

2.4.2. Rheological measurements

Rheological measurements were conducted using a Thermo Fisher HAAKE MARS 60 rheometer (Karlsruhe, Germany) equipped with a cone-plate (60 mm diameter, 1° angle). This equipment was used to investigate the rheological properties of the HA/BSP mixture, including apparent viscosity, viscoelasticity, and temperature-dependent behavior (Doderio et al., 2019). The apparent viscosity was measured as a function of shear rate (ranging from 0.01 to 1000 s^{-1}) at 25 °C (Jin-Long et al., 2016). To characterize the gels, a strain sweep program was initially conducted within a range of 0.1–1000 % strain at 25.0 °C to determine the linear viscoelastic region (LVR) for all mixed gels. Dynamic oscillatory frequency sweep tests, ranging from 0.1 to 100 rad/s, were performed at 500 % strain (within the LVR) to explore the mechanistic response of the gels (Zhao et al., 2024).

The gel underwent an initial temperature sweep from 80 °C to 5.0 °C, conducted at 100.0 % strain and 6.28 rad/s, with a cooling rate of 1 °C/min. Subsequently, the gel was scanned by heating from 5 °C to 80 °C at the same rate.

2.5. Thermal analysis (TG-DTG, DSC)

The thermal properties of the samples were investigated using a simultaneous thermal analyzer (HITACHI STA200, Japan). High-purity nitrogen was used as both purge and protective gas, and measurements were carried out from 30 °C to 1000 °C at a heating rate of 5 °C/min (Dong et al., 2023).

2.6. Scanning electron microscopy (SEM)

After drying, 5 mg of BSP was adhered to a conductive carbon film using double-sided adhesive and placed in the sample chamber of an ion sputtering instrument. A gold coating was then applied for approximately 40 s, followed by observation with a high-resolution scanning

electron microscope (Nova NanoSEM 450, FEI, USA) (Feng & Wang, 2022).

All gels were prepared according to the method mentioned in section 2.4.1, followed by equilibration at room temperature for 12 h. All gels were rapidly frozen by immersion in liquid nitrogen and subsequently freeze-dried in a vacuum freeze dryer. Break the freeze-dried hydrogel manually, take a trace amount of sample and place it directly onto conductive adhesive, then use the Quorum SC7620 sputter coater to gold sputter for 45 s at 10 mA. Subsequently, the samples were imaged using scanning electron microscopy (SEM, TESCAN MIRA LMS), with an accelerating voltage of 3 kV during imaging (Luo et al., 2018).

2.7. Fourier transform infrared spectroscopy (FTIR)

Mix 2 mg of BSP with 200 mg of potassium bromide, and compress the mixture into tablets. A blank control was prepared by compressing potassium bromide powder alone. The samples were then scanned and recorded using FT-IR650 (Liu et al., 2017).

FTIR spectra of freeze-dried samples were obtained by Fourier Transform Infrared Spectrometer (Thermo Scientific Nicolet iS20, USA). In a dry environment, approximately 10 mg of the sample and a suitable amount of dried potassium bromide powder were added to a mortar, thoroughly ground multiple times, and then pressed into a transparent thin pellet using a pellet press machine. The spectral range tested was 400–4000 cm^{-1} with a resolution of 4 cm^{-1} , and the infrared spectrum of the sample was collected (Fimbres-Olivarría et al., 2016).

2.8. Statistical analysis

Statistical analysis and graphing were performed using OriginPro Learning Edition (2024) and Excel 2021. All the measurements were performed with three replicates.

3. Results and discussion

3.1. Structural characterization of BSP-182

3.1.1. Monosaccharide composition and molecular weight determination

BSP was extracted from the tuber of *Bletilla striata*, yielding 20.3 %. Fig. 1 A and Fig. 1B depict the ion chromatograms of BSP and a mixture of 16 sugars. The main monosaccharides in BSP are glucose (16.309 min) and mannose (19.425 min), with a molar ratio of 2.2:7.8 (Fig. 1 A). Table S1 lists the 16 monosaccharide standards. The mannose-to-glucose ratio in BSP can vary with different extraction and purification methods, which consequently affects its physicochemical properties (Chen et al., 2021; Lian-Lin et al., 2023; Wang et al., 2024). BSP exhibited a single, symmetrical, narrow peak on HPLC with a retention time of 26.127 min, confirming its homogeneity (Fig. 1C). The weight-average molecular weight (Mw) and number-average molecular weight (Mn) of BSP were determined to be 182.10 kDa and 169.94 kDa, respectively. The molecular weight standard curve is shown in Fig. S1. We named the novel BSP as BSP-182 based on the Mw. These findings support the dominance of glucose and mannose as the principal monosaccharides in BSP, consistent with previous studies that have characterized BSP as a glucomannan polysaccharide (Wang et al., 2014; Wang et al., 2020).

3.1.2. FTIR analysis of BSP-182

FTIR spectroscopy facilitates the qualitative analysis of polysaccharide functional groups (Tang et al., 2023). The characteristic FTIR spectrum of BSP-182, as shown in Fig. 1D, features an absorption band between 3600 and 3200 cm^{-1} . This band corresponds to the stretching vibrations of the -OH groups, indicative of sugars. Notably, the specific absorption peak at 3375 cm^{-1} is due to the stretching vibrations of O—H, further confirming its sugar-related properties (Zhang et al., 2021). An absorption peak at 2887 cm^{-1} is likely attributed to the C—H stretching vibration. Another peak at 1730 cm^{-1} is possibly due to the

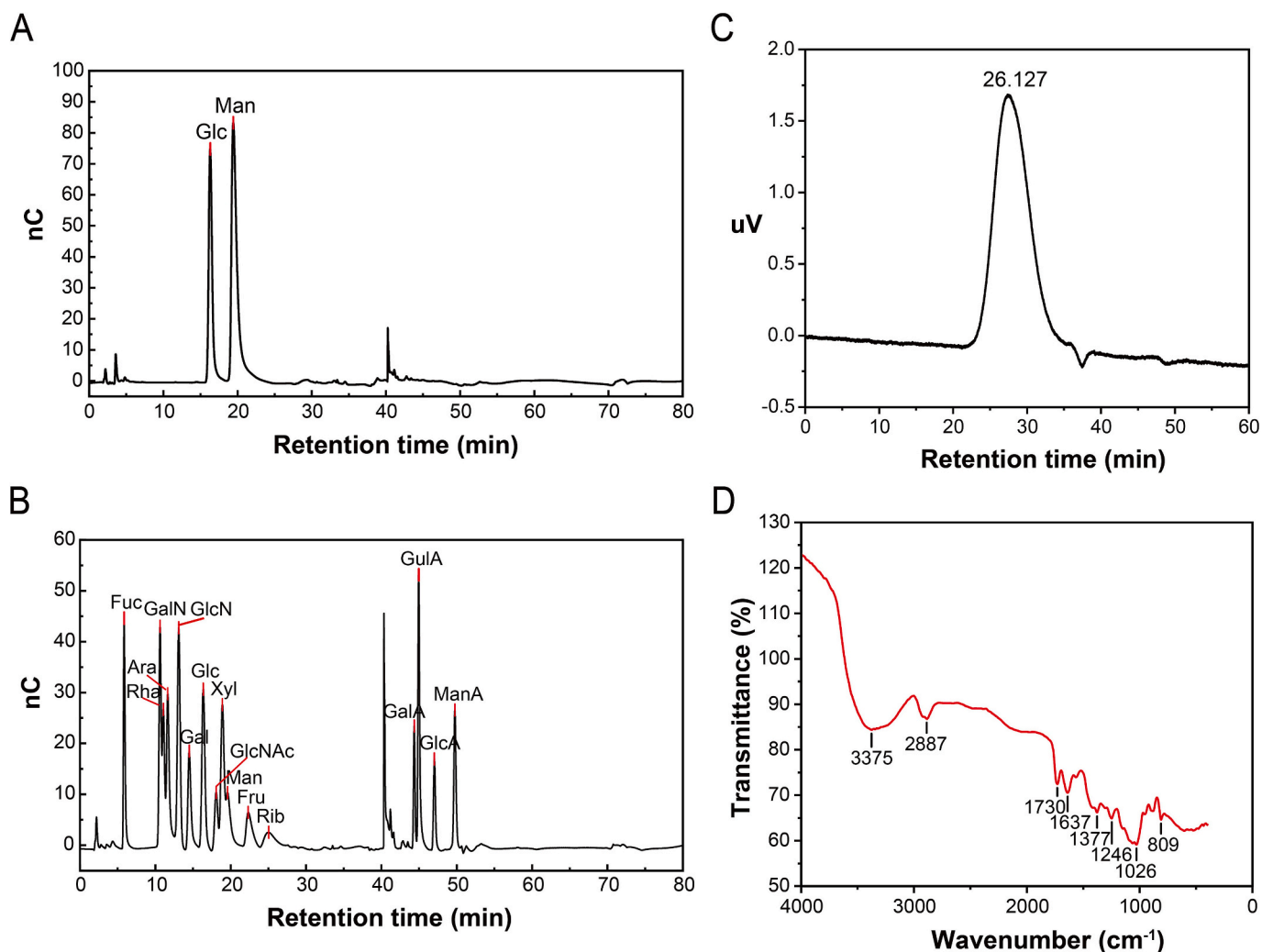


Fig. 1. (A) Monosaccharide composition analysis of BSP; (B) Ion chromatogram of 16 sugars mixed standard; (C) Molecular weight profile of BSP; and (D) FTIR spectra.

C=O stretching vibration. The peak at 1637 cm⁻¹ may be associated with N–H bending vibrations. Additionally, the peak at 1377 cm⁻¹ suggests C=O symmetric stretching vibration. Peaks observed at 1246 cm⁻¹ and 1026 cm⁻¹ could be related to O–H bending vibrations. Finally, the peak at 809 cm⁻¹ is potentially indicative of α -D-galactopyranose (Lin et al., 2022).

3.1.3. Methylation analysis of BSP-182

The polysaccharide was methylated and subsequently analyzed by gas chromatography–mass spectrometry (GC–MS) to determine the types of residues and glycosidic linkages (Ye et al., 2021). As shown in Table 1, BSP-182 contains eight types of glycosidic linkages, with $\rightarrow$ 4)-Manp-(1 $\rightarrow$, $\rightarrow$ 4)-Glc-(1 $\rightarrow$, and Manp-(1 $\rightarrow$ being the most abundant.

The molar ratio of mannose (Man) to glucose (Glc) is approximately 7.8:2.2.

GC–MS chromatogram of the partially methylated alditol acetates (PMAAs) derived from the sample and BSP-182, respectively (Fig. S2 and Fig. S3). Notably, mannose serves as the terminal sugar, comprising 8.7 % of the total, suggesting a branching degree of about 1/11. This indicates a relatively high degree of branching in the polysaccharide. Predominantly, the glycosidic bonds are 1,4-linked, which suggests that the backbone of the polysaccharide primarily consists of 1,4-linked Man and Glc. Branching within the structure occurs at the 2,3,6 positions of the Man residues and the 3,6 positions of the Glc residues. Methylation analysis revealed that the backbone of BSP-182 is primarily composed of 1,4-linked Man and Glc, consistent with the monosaccharide

Table 1
Methylation analysis of BSP-182.

Retention Time (min)	Methylated sugar	Mass fragments (m/z)	Molar ratio	Type of linkage
16.824	2,3,4,6-Me ₄ -Manp	45,71,87,101,117,129,145,161,205	0.087	Manp-(1 $\rightarrow$
21.131	2,3,6-Me ₃ -Manp	45,87,99,101,113,117,129,131,161,173,233	0.658	$\rightarrow$ 4)-Manp-(1 $\rightarrow$
21.36	2,3,6-Me ₃ -Glc	45,87,99,101,113,117,129,131,161,173,233	0.212	$\rightarrow$ 4)-Glc-(1 $\rightarrow$
23.819	2,6-Me ₂ -Glc	45,87,97,117,129,159,185	0.009	$\rightarrow$ 3,4)-Glc-(1 $\rightarrow$
24.132	2,6-Me ₂ -Manp	45,87,97,117,129,159,185	0.008	$\rightarrow$ 3,4)-Manp-(1 $\rightarrow$
24.945	3,6-Me ₂ -Manp	45,87,99,113,129,173,189,233	0.010	$\rightarrow$ 2,4)-Manp-(1 $\rightarrow$
26.016	2,3-Me ₂ -Manp	45,71,85,87,99,101,117,127,159,161,201,261	0.010	$\rightarrow$ 4,6)-Manp-(1 $\rightarrow$
26.312	2,3-Me ₂ -Glc	45,71,85,87,99,101,117,127,159,161,201,261	0.007	$\rightarrow$ 4,6)-Glc-(1 $\rightarrow$

composition analysis.

3.1.4. 1D and 2D NMR of BSP-182

The ^1H NMR spectrum displayed multiple anomeric hydrogen signals in the region of 4.3–5.5 ppm (Fig. 2A). The major signals, identified at 5.43, 5.33, 5.11, 5.03, 4.87, and 4.45 ppm, suggest the presence of various sugar residues and both α and β anomeric configurations. The predominance of the β configuration was determined based on the chemical shifts and signal intensity (Wang et al., 2019). The 3.0–4.2 ppm interval of this ^1H NMR spectrum typically displays proton signals from non-anomeric hydrogens outside the residual sugar ring, exhibiting diverse and complex patterns that align with the general characteristics of polysaccharide hydrogen spectra.

In the ^{13}C NMR spectrum (Fig. 2B), no significant signals corresponding to sugar aldehyde carbonyl carbons were detected within the range of 170–210 ppm, preliminarily confirming the neutrality of the BSP-182. In the anomeric region of the carbon spectrum (90–110 ppm), two principal anomeric carbon signals were observed at 102.50 and 100.13 ppm. The positions of these peaks suggest that the majority of the residual sugars in the BSP-182 predominantly adopt a β -configuration (Silveira et al., 2014); Multiple typical residual sugar carbon peaks were detected in the range of 55–85 ppm. Signals between 58 and 65 ppm correspond to the exocyclic C6 methylene -CH₂- carbons of pyranose type residual sugars, confirming their presence in the monosaccharide composition. This confirmation is further supported by DEPT spectrum analyses, which establish the characteristic correlation of the signals in that region (Fig. S4). Additionally, signals for C2–C5 carbons, outside the anomeric carbons of residual sugars within the sugar rings, were observed in the range of 65–85 ppm, with signal overlap typical of polysaccharide carbon spectra.

The structure of BSP-182 was analyzed using the 2D NMR spectrum, particularly through the HSQC spectrum, which reflects hydrogen-carbon correlations (Fig. 2C). In the anomeric region, multiple sets of H/C correlation signals were identified, with the primary six sets being 5.44/71.42, 5.12/93.81, 4.91/97.72, 4.87/99.05, 4.69/100.11, 4.46/102.50 ppm. Signals at other positions exhibited a lower response, suggesting a lesser abundance of the corresponding residual sugars. By integrating the methylation composition ratios with the relative intensities of the anomeric hydrogen signals in the ^1H spectrum and the HSQC data, preliminary assignments include: 4.69/100.11 ppm with the H1/C1 of a β -1,4-Man residue, 4.46/102.50 ppm with the H1/C1 of a β -1,4-Glc residue, and 5.12/93.81 ppm with the H1/C1 of an α -t-Man residue (Shi et al., 2020; Zhang et al., 2022).

We continued to map the positions of hydrocarbon signals from the three main residues in the polysaccharide structure by analyzing multiple 2D NMR spectra (Fig. 2D). For instance, the anomeric hydrogen at 4.69 ppm was identified as H1 of a β -1,4-Man residue. Neighboring proton correlations in the COSY and NOESY spectra revealed a correlated signal with H1 at 4.06 ppm, thereby establishing the position of H2 at 4.06 ppm. Further analysis using the HSQC spectrum confirmed the H2/C2 signal at 4.06/69.90 ppm, pinpointing the carbon signal for position 2 of this residue at 69.90 ppm. Utilizing the COSY and NOESY data, the related signal at 4.06/3.74 ppm was located, verifying the position of H3 at 3.74 ppm. Subsequent integration with the HSQC spectrum led to the identification of 3.75/71.42 ppm for the ^{13}C spectrum position of C3. Applying this methodology, the positions of H4/C4, H5/C5, and H6/C6 were determined to be at 3.75/76.42 ppm, 3.50/74.90 ppm, and 3.69/60.33 ppm, respectively (Campestrini et al., 2013; Kuang et al., 2020; Zhang et al., 2020). Using the same approach, we further determined the potential attributions of positions for the three main residues (Table 2). In summary, our polysaccharide contains both α and β glycosidic linkages, with the β configuration predominating based on the chemical shifts and relative intensities of the anomeric proton signals.

HMBC spectra are particularly valuable for identifying hydrogen-carbon correlations separated by 2–3 chemical bonds, crucial in

analyzing BSP-182 to determine glycosidic linkages (Fig. 2E). In samples with a high content of β -1,4-Man residues, the HMBC spectra reveal distinctive correlations for the hydrogen at anomeric position: 4.69/70.02, 4.69/76.42. These correlations confirm its association with C2 within the same residue and C4, respectively, thereby establishing the glycosidic linkage as a 1 $\rightarrow$ 4 linkage between β -1,4-Man residues (Sharma & Sharma, 2022).

The NOESY spectrum is instrumental in identifying proton correlation signals of polysaccharide residues that are proximal in space, and it also significantly influences the analysis of glycosidic linkage connections between sugar residues (Fig. 2F). In β -1,4-Man residues with a high composition percentage, the NOESY spectra revealed anomeric hydrogen correlation signals at 4.69/3.49, 4.69/3.74, and 4.69/4.06. These signals correspond to correlations between the anomeric H1 and H5, H3, H2 within the same ring, and H4 of the same residue, respectively, confirming the glycosidic bond of the 1 $\rightarrow$ 4 linkage in β -1,4-Man. Similarly, in β -1,4-Glc residues, correlations at 4.46/3.61 and 4.46/3.73 were identified, indicating 1 $\rightarrow$ 4 glycosidic bonds within the residue itself and between β -1,4-Glc and β -1,4-Man, respectively. Additionally, a signal at 5.12/3.75 in α -t-Man residues suggests a 1 $\rightarrow$ 4 glycosidic linkage with β -1,4-Man.

The main structure of BSP-182 can be delineated based on the monosaccharide composition ratios and methylation analysis results, utilizing the structures of the three primary sugar residues and considering the types of other remaining residues (Fig. 2G). ("X" denotes the set of sugar residues present in smaller quantities: $\rightarrow$ 3,4)-Glc-(1 $\rightarrow$, $\rightarrow$ 3,4)-Manp-(1 $\rightarrow$, $\rightarrow$ 2,4)-Manp-(1 $\rightarrow$, $\rightarrow$ 4,6)-Manp-(1 $\rightarrow$, $\rightarrow$ 4,6)-Glc-(1 $\rightarrow$).

3.1.5. Analysis of BSP-182 by SEM

The morphology and microstructure of BSP-182 were characterized using Scanning Electron Microscopy (SEM). Under 200 $\times$ magnification, the sample presented as thin layers, exhibiting a thin, incomplete, and irregularly shaped structure with irregular stacking and noticeable gaps (Fig. 2H). These morphological characteristics of BSP-182 are common, likely due to the strong attraction between the functional groups on the polysaccharide surfaces, which results in the aggregation of polysaccharide chains (Li et al., 2017). When magnified to 500 $\times$, the edges of the thin layers were clearly defined, and the fracture surfaces were evident, characterized by a smooth surface. At higher magnifications of 1000 $\times$ and 2000 $\times$, filamentous branches were observed at the layer edges, displaying a distinct branch structure with neat and smooth edges (Fig. S5).

3.2. Synergistic gelation of BSP-182 and HA

3.2.1. Rheology of HA/BSP-182 mixtures

Rheological properties, including apparent viscosity, viscoelasticity, and temperature-dependent behavior, were determined for the pure BSP-182 solution and HA/BSP mixtures using a rheometer. The rheological properties of HA/BSP-182 mixtures indicate that the apparent viscosity of all HA/BSP-182 mixtures varies with shear rate (Fig. 3A). The BSP-182 solution exhibits Newtonian behavior at high shear rates, whereas mixing with HA increases the viscosity of the solution, demonstrating significant shear-thinning behavior. These results are highly consistent with other composite gel rheological characteristics of BSP (Cui et al., 2017; Xiang et al., 2022). Shear-thinning is a typical behavior of polysaccharide hydrogels with high molecular weight and polymer structure, which contributes to material preparation, shaping, and structural improvement (W. Feng et al., 2022; Tang et al., 2022), especially in beauty applications, oral liquids, food additives, which require a finer texture (Chao et al., 2024). Shear-thinning behavior may be related to several factors, including polymer structure (Li et al., 2024; Liu et al., 2023), additives (Kongjaroen et al., 2022; Maas et al., 2013), external conditions (Kapadia et al., 2022; Silva et al., 2017), and polymer concentration (Martín-Alfonso et al., 2018; Torres et al., 2014).

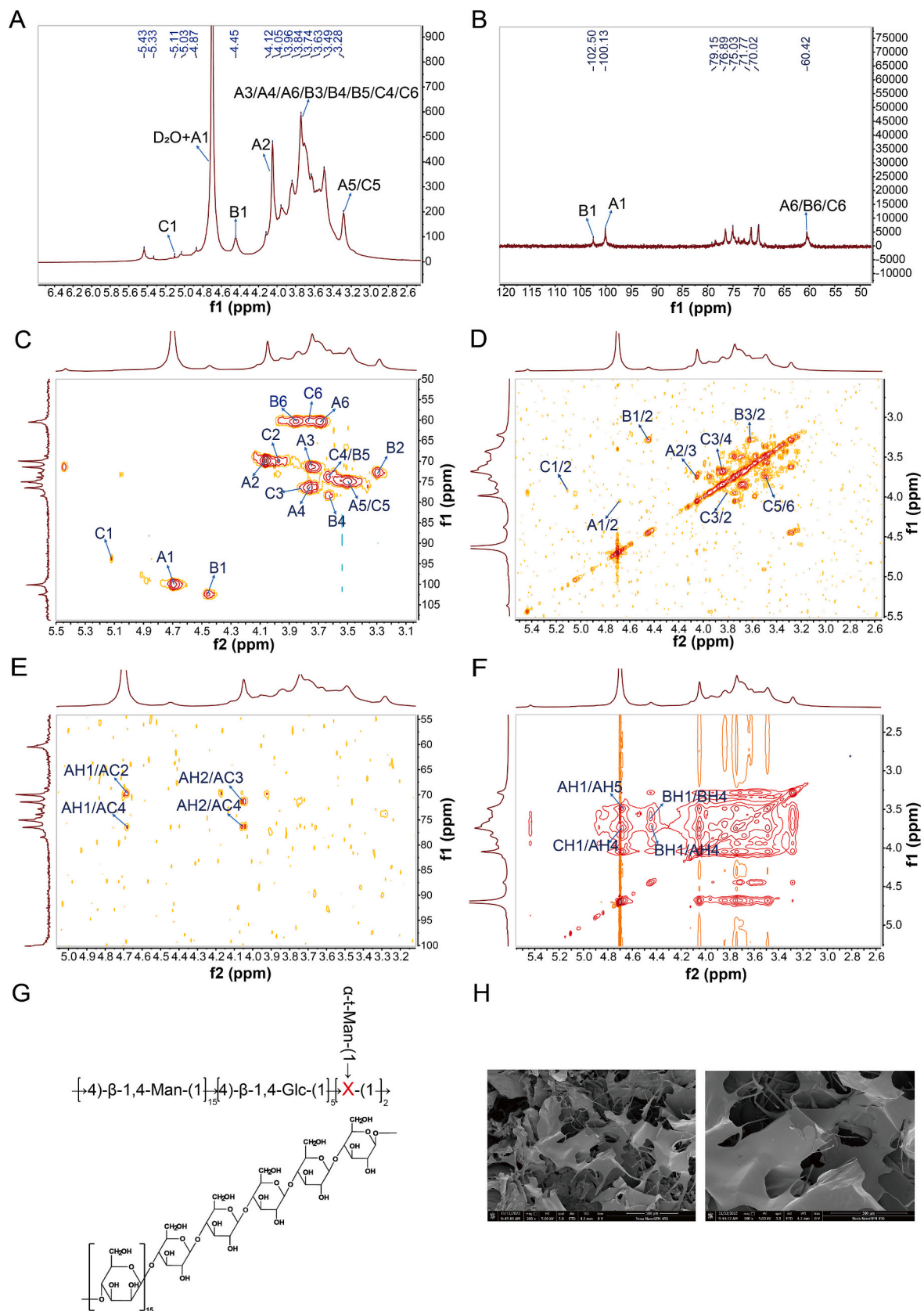


Fig. 2. The 1D NMR and 2D NMR spectra of BSP-182 (A) ^1H , (B) ^{13}C , (C) HSQC, (D) HMBC, (E) COSY, (F) NOESY, (G) Predicted structure of the repeating unit of BSP-182 and (H) SEM images of BSP-182 at 200 $\times$ and 500 $\times$ magnification.

Table 2

Main residual sugar data of BSP-182.

Residual sugar (code)	H1/C1	H2/C2	H3/C3	H4/C4	H5/C5	H6/C6
β -1,4-Man (A)	4.69/100.11	4.06/69.90	3.75/71.42	3.75/76.42	3.50/74.90	3.69/60.33
β -1,4-Glc (B)	4.46/102.50	3.28/72.72	3.62/73.81	3.63/78.37	3.68/71.77	3.86/60.33
α -t-Man (C)	5.12/93.81	3.91/70.02	3.82/76.49	3.67/71.77	3.48/74.90	3.75/60.42

Additionally, the mixture solution exhibits the highest viscosity when the HA/BSP-182 ratio is 7:3.

The storage modulus (G') and loss modulus (G'') increase as the frequency rises from 1 rad/s to 100 rad/s (Fig. 3B). Throughout the frequency sweep test, the material behaves like a viscoelastic liquid below the crossover frequency ($G'' > G'$), with the crossover occurring at very high frequencies. This behavior may be due to an increasing trend of entangled polymer collisions at higher frequencies, leading to a faster increase in the storage modulus (Zad Bagher Seighalani et al., 2021). Additionally, the modulus values are highest at an HA ratio of 7:3, indicating better viscoelastic properties of the hydrogel at this ratio (do Amaral Sobral et al., 2022).

Strain sweep tests were conducted to identify the linear viscoelastic

region (LVR) of the hybrid HA-BSP-182 system (Fig. S6). The results demonstrate that across all strain ranges, the samples exhibit wide LVRs. Additionally, all proportions of HA/BSP-182 show no yield points and flow without the need for additional stress, facilitating easy liquid flow. When the strain amplitude is less than 10 %, the G' and G'' of mixtures with different mass ratios of HA and BSP-182 both plateau. However, when the strain exceeds 10 %, molecular entanglements may be disrupted. For a ratio of HA to BSP-182 of 7:3, the hydrogel exhibits the highest values of G'' and G' values, indicating the strongest resistance to mechanical deformation (Wei et al., 2021).

We performed a dynamic oscillatory frequency sweep test from 0.1 to 100 rad/s was conducted at 500 % strain (within the LVR) to investigate the mechanical response of the gel. An initial temperature sweep from 80 °C to 5.0 °C was performed on the gel at 100.0 % strain and 6.28 rad/s with a cooling rate of 1 °C/min. Subsequently, the gel was scanned from 5 °C to 80 °C upon heating at the same rate. During the cooling process from 80 °C to 5 °C, both the G'' and G' increase, due to the reduction in molecular mobility, with G'' remaining higher than G' (Fig. 3C). The HA/BSP-182 mixture maintains a colloidal state throughout. When the temperature drops below 40 °C, the rapid increase in the viscous modulus indicates a higher viscosity at this point. Upon heating, the viscous modulus of the hydrogel decreases with rising temperature, leveling off at 40 °C (Fig. 3D). This demonstrates good thermoreversibility of both G' and G'' in the hydrogel. Throughout the

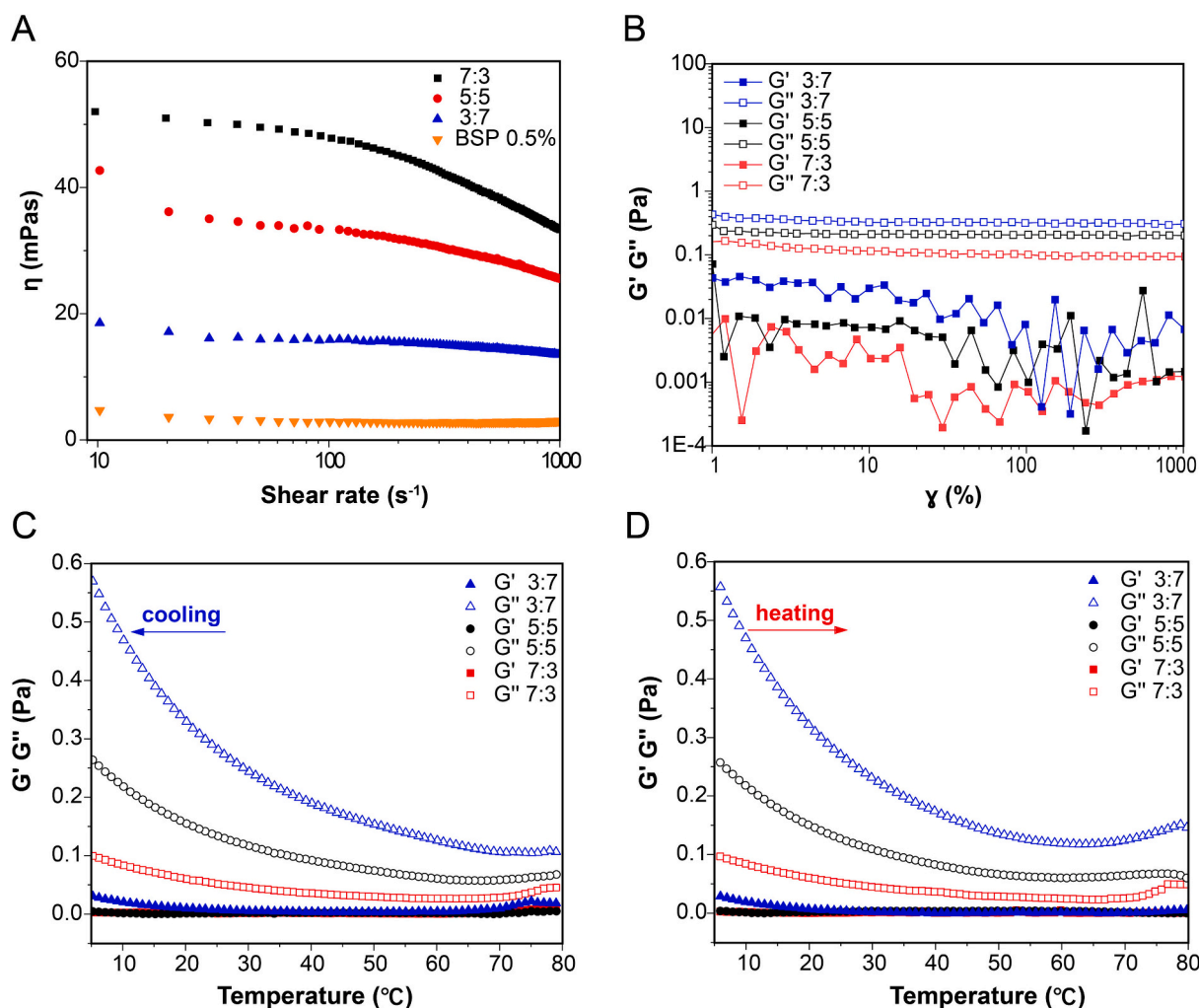


Fig. 3. Rheology of HA/BSP-182 mixtures. (A) Apparent viscosity of BSP-182 (0.5 %, w/v), HA/BSP-182 mixtures with different ratios (HA: BSP-182 = 7:3, HA: BSP-182 = 5:5 and HA: BSP-182 = 3:7) at 25.0 °C. (B) Changes in G' and G'' of HA/BSP-182 mixtures with different ratios (HA: BSP-182 = 7:3, HA: BSP-182 = 5:5 and HA: BSP-182 = 3:7) at 25.0 °C and 500 % strain. (C) and (D) Variation of G' , G'' (at 6.28 rad/s and 100 % strain) during cooling (80.0–5.0 °C) and reheating (5.0–80.0 °C) for HA and BSP-182 at different ratios (7:3, 5:5 and 3:7).

heating and cooling processes, the modulus values of the HA/BSP-182 = 7:3 mixture consistently remain the highest. These results indicate that compared to the pure BSP-182 solution, the HA/BSP-182 mixed solution forms a stronger colloid through synergistic effects, suggesting that HA may serve as an effective coagulant for BSP.

3.2.2. Thermal analysis of HA/BSP-182

The samples underwent TG/DSC analysis utilizing a simultaneous thermal analyzer. The measurements were executed in a high-purity nitrogen atmosphere, ranging from 30 °C to 1000 °C at a heating rate of 5 °C/min. Data was collected from thermogravimetric (TG), derivative thermogravimetric (DTG), and differential scanning calorimetric (DSC) analyses.

The thermal stability and degradation of HA/BSP-182 hydrogels were analyzed using TG/DTG and DSC curves (Fig. 4). Fig. 4 A presents the TG/DTG profiles, where weight loss was measured to record thermogravimetric data. All weight loss processes between 30 and 200 °C were recorded. The five profiles exhibit similar decomposition stages, where the weight loss is attributed to the evaporation of free water, i.e., water bound by weak physical interactions such as hydrogen bonds (Fontana et al., 2022). The hydrogels demonstrate lower thermal stability compared to the pure solutions, likely due to a higher abundance of hydrogen bonds, requiring less energy for breakage. Fig. 4B shows the DSC curves, revealing a significant endothermic peak for the hydrogels with different ratios in the lower temperature range (approximately 50–150 °C). This difference is closely related to the hydrogel composition. The HA/BSP-182 (7:3) hydrogel exhibits a higher phase transition temperature and greater thermal stability, suggesting a higher internal crosslinking density, restricting water molecule migration and desorption. This higher crosslinking density likely enhances the structural integrity of the hydrogel, increasing its resistance to thermally induced phase transitions.

3.2.3. HA/BSP-182 SEM analysis

SEM was used to visualize the hydrogel network structure. Pure BSP-182 exhibited a lamellar and smooth structure, whereas HA displayed a fibrous structure with entangled molecular chains (Fig. 5).

In contrast, the mixture of HA and BSP-182 exhibited a more pronounced honeycomb structure. This could be attributed to the strong synergistic effect between HA and BSP-182 in the co-hydrocolloid, which led to a reduction in the size of the pore network in the mixed colloid. The outer surface of all samples appeared rough, with distinct

grooves and pores, helping to maintain the spatial structure and enabling the infiltration and retention of moisture (Farasati Far et al., 2023). It is worth noting that a more regular and denser network was observed at the ratio of HA:BSP-182 = 7:3, characterized by more numerous and smaller pores. This ratio corresponded to the maximum value of hydrogen bond connections (Khoobbakht et al., 2024).

3.2.4. FTIR spectra of HA/BSP-182 hydrogels

The interaction between HA and BSP was confirmed by FTIR (Fig. 6). Fig. S7 provides the annotated infrared spectra. The infrared characteristics of the BSP sample are similar to those of HA, with absorption peaks at 3389 cm^{-1} , 2892 cm^{-1} , 1643 cm^{-1} and 1033 cm^{-1} originating from O—H, C—H, C=O and C—O structures, respectively, likely due to both being polysaccharides. Notably, a unique ester C=O stretching vibration absorption peak was observed at 1732 cm^{-1} , and the absorption peak at 895 cm^{-1} is related to the β -glycosidic bond, indicating that the glycosidic bonds in this component are mainly in the β -configuration (Khoobbakht et al., 2024).

After crosslinking BSP-182 with HA, the infrared spectrum of the sample exhibits characteristic peaks of HA. In the hydrogel with a ratio of HA:BSP = 7:3, absorption peaks corresponding to the C=O stretching vibration in BSP are observed at 1726 cm^{-1} and 1412 cm^{-1} . The intensities of these C=O stretching vibration peaks decrease gradually with lower BSP ratios in the other two hydrogels (Hou et al., 2023a). Additionally, as the BSP ratio decreases, the shape of the O—H stretching vibration absorption peak changes, with the hydroxyl peak intensity in hydrogels being higher than in the pure BSP solution. This indicates that BSP has been encapsulated by HA and forms complex hydrogen bonding interactions through hydroxyl groups and other structures.

4. Conclusions

In this study, a novel water-soluble polysaccharide, designated BSP-182, was isolated and purified from the tubers of *Bletilla striata*. Structural analysis revealed that BSP-182 is composed of a backbone consisting of $\rightarrow 3,4$ -GlcP-(1 $\rightarrow$, $\rightarrow 3,4$ -Manp-(1 $\rightarrow$, $\rightarrow 2,4$ -Manp-(1 $\rightarrow$, $\rightarrow 4,6$ -Manp-(1 $\rightarrow$, and $\rightarrow 4,6$ -GlcP-(1 $\rightarrow$ linkages. BSP-182 interacts with hyaluronic acid (HA) via hydrogen bonding to form hydrogels, which exhibit superior rheological properties and microstructure compared to the pure BSP-182 solution. All HA/BSP-182 hydrogels displayed shear-thinning behavior and thermoreversibility at 40 °C. The optimal

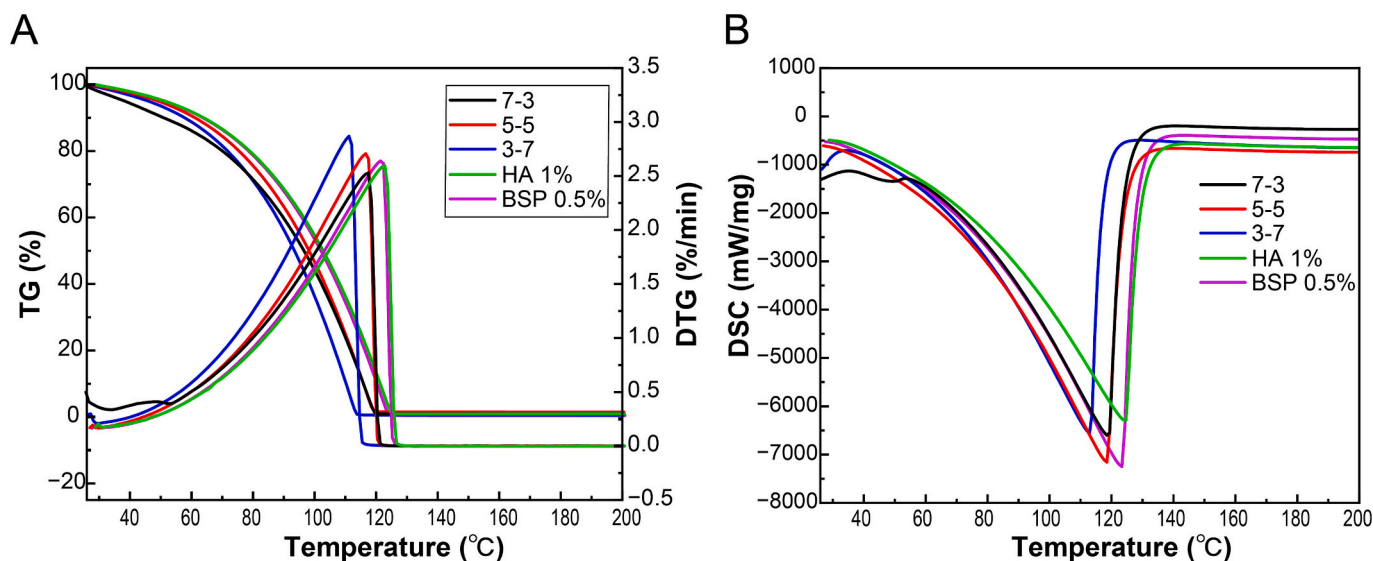


Fig. 4. TG/DTG-DSC curves of HA/BSP-182 mixtures. (A) TG/DTG curves, and (B) DSC curves of HA/BSP-182 (7:3, 5:5, and 3:7), BSP-182 (0.5 % w/v), and HA (1 % w/v).

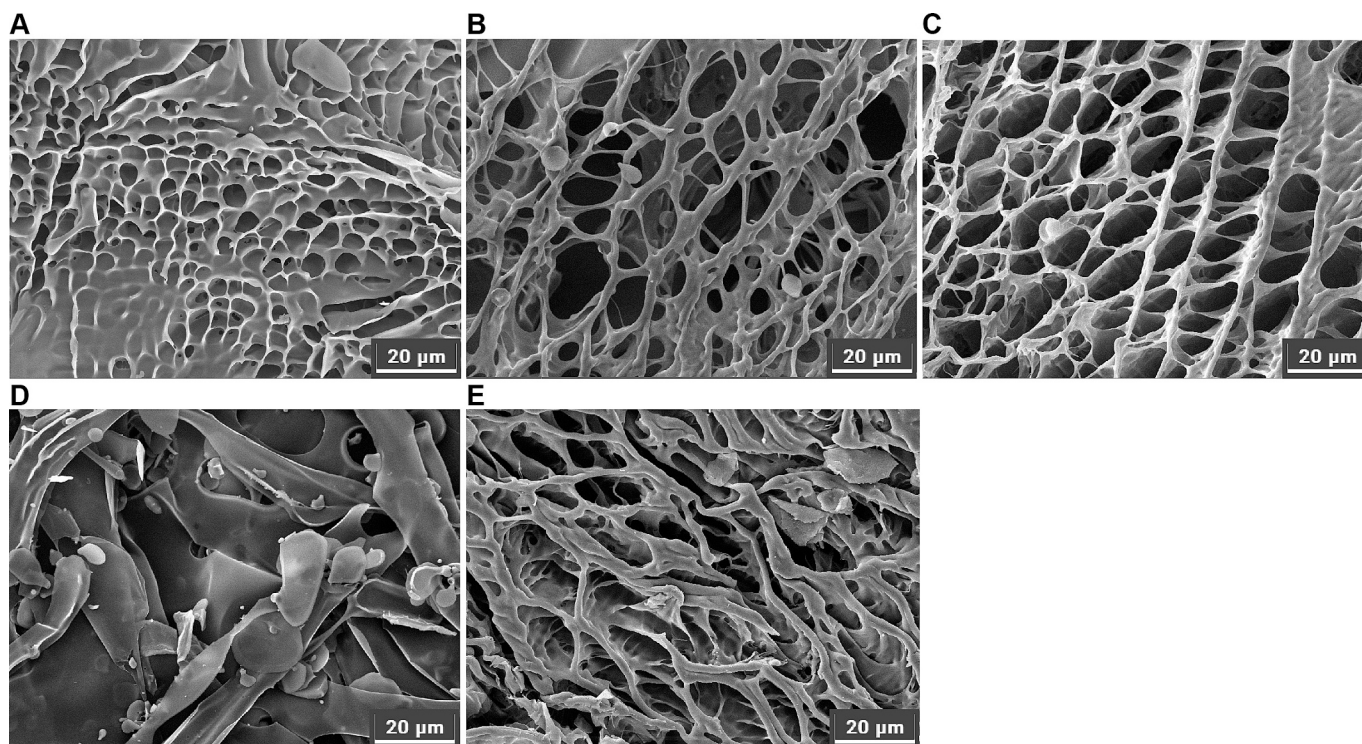


Fig. 5. Photomicrographs of HA/BSP-182 mixtures captured using SEM at 500× magnification. (A) HA: BSP-182 = 7:3; (B) HA: BSP-182 = 5:5; (C) HA: BSP-182 = 3:7; (D) BSP-182; (E) HA.

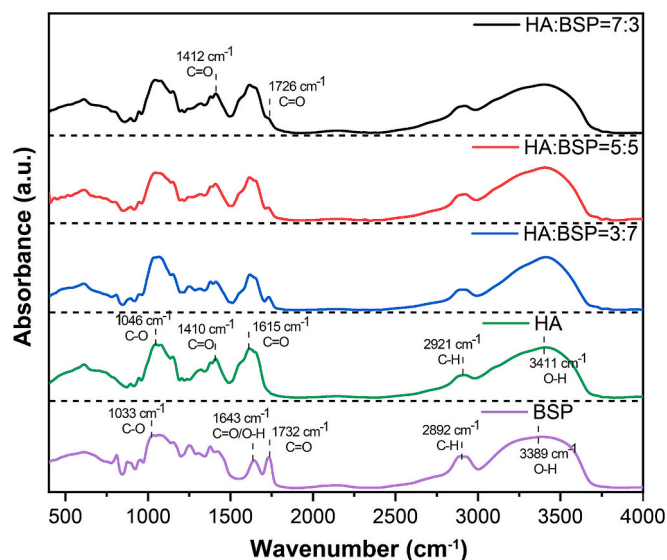


Fig. 6. FTIR spectra of HA/BSP-182 hydrogels.

storage modulus (G') and loss modulus (G'') were achieved with a HA/BSP-182 ratio of 7:3, suggesting an increase in viscoelasticity and a heightened resistance to mechanical deformation, characteristics that are conducive to safe swallowing. It is noteworthy that HA/BSP-182 hydrogels retain the nutritional properties of BSP-182, highlighting their potential as functional food ingredients. This finding provides a theoretical foundation for the development of easy-to-consume hydrogel-based foods.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodchem.2024.142359>.

Funding

We thank the editors and reviewers for their careful reading and helpful comments on this manuscript. This work was supported by the National Natural Science Foundation of China (Grant No. 82304680), and Anhui Provincial Natural Science Foundation (2308085QH295, 2308085MH261), and Natural Science Research Project of Anhui Educational Committee (Grant Number: 2023AH040259), and the Talent Scientific Research Start-up Foundation, Wannan Medical College (Grant Number: YR20230110); Anhui Provincial Department of Education Young Backbone Teachers Overseas Visiting and Training Funding Program (JWFX2023033), and Anhui Province Outstanding Young Talent Youth Scholar Program, and Anhui province university science and engineering teachers to the enterprise practice plan project (2024jsqygz60).

CRediT authorship contribution statement

Didi Ma: Writing – original draft, Formal analysis. **Zhenzhen Zhao:** Writing – original draft, Formal analysis. **Yujing Wen:** Formal analysis. **Juan Zhou:** Formal analysis. **Wenhao Zhou:** Formal analysis. **Jian Mao:** Formal analysis. **Kun Lv:** Data curation. **Yunpeng Cao:** Data curation. **Lan Jiang:** Writing – review & editing, Funding acquisition, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- do Amaral Sobral, P. J., Gebremariam, G., Drudi, F., Pinheiro, A. C. D. A. S., Romani, S., Rocculi, P., & Rosa, M. D. (2022). Rheological and viscoelastic properties of chitosan solutions prepared with different chitosan or acetic acid concentrations. *Foods*, 11 (17).
- Avadhani, K. S., Manikkath, J., Tiwari, M., Chandrasekhar, M., Godavarthi, A., Vidya, S. M., ... Mutalik, S. (2017). Skin delivery of epigallocatechin-3-gallate (EGCG) and hyaluronic acid loaded nano-transfersomes for antioxidant and anti-aging effects in UV radiation induced skin damage. *Drug Delivery*, 24(1), 61–74.
- Bao, Z., Xian, C., Yuan, Q., Liu, G., & Jun, W. (2019). Natural polymer-based hydrogels with enhanced mechanical performances: Preparation, structure, and property. *Advanced Healthcare Materials*, 8(17), Article e1900670.
- Campestrini, L. H., Silveira, J. L. M., Duarte, M. E. R., Koop, H. S., & Nosedá, M. D. (2013). NMR and rheological study of Aloe barbadensis partially acetylated glucomannan. *Carbohydrate Polymers*, 94(1), 511–519.
- Cao, W., Jiangping, W., Zhao, X., Li, Z., Jie, Y., Shao, T., ... Han, J. (2024). Structural elucidation of an active polysaccharide from Radix Puerariae lobatae and its protection against acute alcoholic liver disease. *Carbohydrate Polymers*, 325, Article 121565.
- Chao, E., Li, J., Duan, Z., & Fan, L. (2024). Bigels as emerging biphasic systems: Properties, applications, and prospects in the food industry. *Food Hydrocolloids*, 110089.
- Chen, H., Zeng, J., Wang, B., Cheng, Z., Jun, X., Gao, W., & Chen, K. (2021). Structural characterization and antioxidant activities of Bletilla striata polysaccharide extracted by different methods. *Carbohydrate Polymers*, 266, Article 118149.
- Chen, Q., Yanyan, W., Cao, L., Zhao, Y., Xiang, H., Lin, H., & Wang, Y. (2024). Preparation and characterization of fish-derived protein gel as a potential dysphagia food: Co-mingling effects of inulin and konjac glucomannan mixtures. *Food Chemistry*, 460, Article 140742.
- Chen, Z., Cheng, L., He, Y., & Wei, X. (2018). Extraction, characterization, utilization as wound dressing and drug delivery of Bletilla striata polysaccharide: A review. *International Journal of Biological Macromolecules*, 120(Pt B), 2076–2085.
- Cui, X., Zhang, X., Yang, Y., Wang, C., Zhang, C., & Peng, G. (2017). Preparation and evaluation of novel hydrogel based on polysaccharide isolated from Bletilla striata. *Pharmaceutical Development and Technology*, 22(8), 1001–1011.
- Delin, X., Pan, Y., & Chen, J. (2019). Chemical constituents, pharmacologic properties, and clinical applications of Bletilla striata. *Frontiers in Pharmacology*, 10, 1168.
- Doderio, A., Williams, R., Gagliardi, S., Vicini, S., Alloisio, M., & Castellano, M. (2019). A micro-rheological and rheological study of biopolymers solutions: Hyaluronic acid. *Carbohydrate Polymers*, 203, 349–355.
- Dong, R., Tian, J., Huang, Z., Qiang, Y., Xie, J., Li, B., ... Chen, Y. (2023). Intermolecular binding of blueberry anthocyanins with water-soluble polysaccharides: Enhancing their thermostability and antioxidant abilities. *Food Chemistry*, 410, Article 135375.
- Fan, Y., Hong, R., Sun, X., Luo, Q., Wei, H., Chen, Y., ... Wan, J. (2024). Gastric acid-responsive deformable sodium alginate/Bletilla striata polysaccharide in situ gel for the protection and treatment of alcohol-induced peptic ulcers. *International Journal of Biological Macromolecules*, 258(Pt 1), Article 128815.
- Far, B. F., Naimi-Jamal, M. R., Jahanbakhshi, M., Rostamani, H., Karimi, M., & Keihankhadiv, S. (2023). Synthesis and characterization of chitosan/collagen/polycaprolactone hydrogel films with enhanced biocompatibility and hydrophilicity for artificial tendon applications. *International Journal of Biological Macromolecules*, 253, Article 127448.
- Feng, W., & Wang, Z. (2022). Shear-thinning and self-healing chitosan-graphene oxide hydrogel for hemostasis and wound healing. *Carbohydrate Polymers*, 294, Article 119824.
- Feng, Y., Qiu, Y., Duan, Y., He, Y., Xiang, H., Sun, W., ... Ma, H. (2022). Characterization, antioxidant, antineoplastic and immune activities of selenium modified Sagittaria sagittifolia L. polysaccharides. *Food Research International (Ottawa Ont.)*, 153, Article 110913.
- Fimbras-Olivarría, D., López-Elías, J. A., Carvajal-Millán, E., Márquez-Escalante, J. A., Martínez-Córdova, L. R., Miranda-Baeza, A., ... Brown-Bojórquez, F. (2016). Navicula sp. sulfated polysaccharide gels induced by Fe(III): Rheology and microstructure. *International Journal of Molecular Sciences*, 17(8).
- Fontana, R., Marconi, P. C. R., Caputo, A., & Gavalyan, V. B. (2022). Novel chitosan-based Schiff Base compounds: Chemical characterization and antimicrobial activity. *Molecules (Basel, Switzerland)*, 27 (9).
- He, X., Liu, L., Fangli, G., Huang, R., Liu, L., Nian, Y., ... Song, C. (2024). Exploration of the anti-inflammatory, analgesic, and wound healing activities of Bletilla Striata polysaccharide. *International Journal of Biological Macromolecules*, 261(Pt 2), Article 129874.
- He, X., Wang, X., Fang, J., Zhao, Z., Huang, L., Guo, H., & Zheng, X. (2017). Bletilla striata: Medicinal uses, phytochemistry and pharmacological activities. *Journal of Ethnopharmacology*, 195, 20–38.
- Hosseini, M. S., Kamali, B., & Nabid, M. R. (2022). Multilayered mucoadhesive hydrogel films based on Ocimum basilicum seed mucilage/thiolated alginate/dopamine-modified hyaluronic acid and PDA coating for sublingual administration of nystatin. *International Journal of Biological Macromolecules*, 203, 93–104.
- Hou, Y., Zhao, J., Yin, J., & Nie, S. (2023a). Structural properties of Bletilla striata polysaccharide and the synergistic gelation of polysaccharide and xanthan gum. *Food Hydrocolloids*, 142.
- Hou, Y., Zhao, J., Yin, J., & Nie, S. (2023b). Structural properties of Bletilla striata polysaccharide and the synergistic gelation of polysaccharide and xanthan gum. *Food Hydrocolloids*, 142, Article 108843.
- Huang, Y., Shi, F., Liming Wang, Y., Yang, B. M., Khan, K.-L. C., & Liu, Y. (2019). Preparation and evaluation of Bletilla striata polysaccharide/carboxymethyl chitosan/Carbomer 940 hydrogel for wound healing. *International Journal of Biological Macromolecules*, 132, 729–737.
- Jiang, L., Lin, M., Wang, H., Song, H., Zhang, L., Huang, Q., ... Cao, Y. (2022). Haplotype-resolved genome assembly of Bletilla striata (Thunb.) Reichb.F. To elucidate medicinal value. *The Plant Journal : For Cell and Molecular Biology*, 111(5), 1340–1353.
- Jin-Long, X., Zhang, J.-C., Liu, Y., Sun, H.-J., & Wang, J.-H. (2016). Rheological properties of a polysaccharide from floral mushrooms cultivated in Huangshan Mountain. *Carbohydrate Polymers*, 139, 43–49.
- Kapadia, W., Qin, N., Zhao, P., Phan, C.-M., Haines, L., Jones, L., & Ren, C. L. (2022). Shear-thinning and temperature-dependent viscosity relationships of contemporary ocular lubricants. *Translational Vision Science & Technology*, 11(3), 1.
- Khoobakht, F., Khorshidi, S., Bahmanyar, F., Hosseini, S. M., Aminikhah, N., Farhoodi, M., & Mirmoghtadaie, L. (2024). Modification of mechanical, rheological and structural properties of agar hydrogel using xanthan and locust bean gum. *Food Hydrocolloids*, 147.
- Kongaroen, A., Methacanon, P., Seetapan, N., Fuongfuchat, A., Gamonpilas, C., & Nishinari, K. (2022). Effects of dispersing media on the shear and extensional rheology of xanthan gum and guar gum-based thickeners used for dysphagia management. *Food Hydrocolloids*, 132, Article 107857.
- Kuang, M.-T., Li, J.-Y., Yang, X.-B., Yang, L., Jing-Yue, X., Yan, S., ... Zhou, J. (2020). Structural characterization and hypoglycemic effect via stimulating glucagon-like peptide-1 secretion of two polysaccharides from Dendrobium officinale. *Carbohydrate Polymers*, 241, Article 116326.
- Li, L., Jing, J., Yang, S., Fang, S., Liu, W., Wang, C., ... Yang, C. (2023). Bletilla striata polysaccharide nanoparticles improved the therapeutic efficacy of omeprazole on the rat gastric ulcer induced by ethanol. *Molecular Pharmaceutics*, 20(4), 1996–2008.
- Li, Q., Wang, W., Zhu, Y., Chen, Y., Zhang, W., Ping, Y., ... Xiangyang, W. (2017). Structural elucidation and antioxidant activity of a novel se-polysaccharide from sen-enriched Grifolia frondosa. *Carbohydrate Polymers*, 161, 42–52.
- Li, X., Zhao, B., Zou, Y., Guo, G., Li, J., Sheng, J., ... Luo, J. (2024). Structure, rheology and stability of walnut oleogels structured by cellulose nanofiber of different lengths. *Food Hydrocolloids*, 154, Article 110148.
- Lian-Lin, S., Li, X., Guo, Z.-J., Xiao, X.-Y., Chen, P., Zhang, J.-B., ... Tu-Lin, L. (2023). Effects of different steaming times on the composition, structure and immune activity of Polygonatum polysaccharide. *Journal of Ethnopharmacology*, 310, Article 116351.
- Lin, P., Chen, L., Huang, X., Xiao, F., Lei, F., Jing, D., ... Yunkun, W. (2022). Structural characteristics of polysaccharide GP2a in Gardenia jasminoides and its immunomodulatory effect on macrophages. *International Journal of Molecular Sciences*, 23(19).
- Lin, Y., Jingchen, X., Dong, Y., Wang, Y., Chenhao, Y., Li, Y., ... Peng, Q. (2023). Drug-free and non-crosslinked chitosan/hyaluronic acid hybrid hydrogel for synergistic healing of infected diabetic wounds. *Carbohydrate Polymers*, 314, Article 120962.
- Liu, J., Shang, F., Yang, Z., Mingyi, W., & Zhao, J. (2017). Structural analysis of a homogeneous polysaccharide from Achatina fulica. *International Journal of Biological Macromolecules*, 98, 786–792.
- Liu, J., Moyuan, Q., Wang, C., Xue, Y., Huang, H., Chen, Q., ... Jiang, X. (2022). A dual-cross-linked hydrogel patch for promoting diabetic wound healing. *Small (Weinheim an Der Bergstrasse, Germany)*, 18(17), Article e2106172.
- Liu, L., Liu, Y., Li, J., Guocheng, D., & Chen, J. (2011). Microbial production of hyaluronic acid: Current state, challenges, and perspectives. *Microbial Cell Factories*, 10, 99.
- Liu, S., Zhang, J., Hutchings, L. R., Peng, L., Huang, X., & Huang, Q. (2023). The “solvent” effect of short arms on linear and nonlinear shear rheology of entangled asymmetric star polymers. *Polymer*, 281, Article 126125.
- Liuying, H., Nomura, S., Sato, Y., Takagi, K., Ishii, T., Honma, Y., ... Muto, J. (2022). Anti-inflammatory effects of differential molecular weight hyaluronic acids on UVB-induced calprotectin-mediated keratinocyte inflammation. *Journal of Dermatological Science*, 107(1), 24–31.
- Luo, P., Nie, M., Wen, H., Wenyan, X., Fan, L., & Cao, Q. (2018). Preparation and characterization of carboxymethyl chitosan sulfate/oxidized konjac glucomannan hydrogels. *International Journal of Biological Macromolecules*, 113, 1024–1031.
- Ma, Z., Yang, X., Ma, J., Lv, J., He, J., Jia, D., ... Zeng, R. (2021). Development of the mussel-inspired pH-responsive hydrogel based on Bletilla striata polysaccharide with enhanced adhesiveness and antioxidant properties. *Colloids and Surfaces. B, Biointerfaces*, 208, Article 112066.
- Maas, M., Bodnar, P. M., Hess, U., Treccani, L., & Rezwani, K. (2013). Towards the synthesis of hydroxyapatite/protein scaffolds with controlled porosities: Bulk and interfacial shear rheology of a hydroxyapatite suspension with protein additives. *Journal of Colloid and Interface Science*, 407, 529–535.
- Martín-Alfonso, J. E., Cuadri, A. A., Berta, M., & Stading, M. (2018). Relation between concentration and shear-extensional rheology properties of xanthan and guar gum solutions. *Carbohydrate Polymers*, 181, 63–70.
- Mzoughi, Z., Abdelhamid, A., Rihouey, C., Le Cerf, D., Bouraoui, A., & Majdoub, H. (2018). Optimized extraction of pectin-like polysaccharide from Suaeda frutescens leaves: Characterization, antioxidant, anti-inflammatory and analgesic activities. *Carbohydrate Polymers*, 185, 127–137.
- Ni, C., Zhang, Z., Wang, Y., Zhang, Z., Guo, X., & Lv, H. (2023). Hyaluronic acid and HA-modified cationic liposomes for promoting skin penetration and retention. *Journal of Controlled Release : Official Journal of the Controlled Release Society*, 357, 432–443.
- Niu, X., Jiabao, Y., Huang, Q., Jinjin, Y., Yang, Y., Song, H., ... Li, W. (2022). Immunoenhancement activity of Bletilla striata polysaccharide through MAPK and NF-κB signalling pathways in vivo and in vitro. *Autoimmunity*, 55(8), 650–660.

- Peng, S., Pengfei, G., Mao, N., Lin, Y., Zhu, T., He, J., ... Wang, D. (2024). Structural characterization and in vitro anti-inflammatory activity of polysaccharides isolated from the fruits of *Rosa laevigata*. *International Journal of Molecular Sciences*, 25(4).
- Qiu, M., Zhong, G., Zhang, J., Hou, Y., Duan, Y., Guo, P., ... Yan, Q. (2024). Biocompatible and biodegradable *Bletilla striata* polysaccharides hydrogels crosslinked by BDDE for wound healing through the regulating of macrophage polarization. *International Journal of Biological Macromolecules*, 254(Pt 3), Article 128015.
- Ren, K., Wan, H., Kaper, H. J., & Sharma, P. K. (2022). Dopamine-conjugated hyaluronic acid delivered via intra-articular injection provides articular cartilage lubrication and protection. *Journal of Colloid and Interface Science*, 619, 207–218.
- Seighalani, F. Z. B., McMahon, D. J., & Sharma, P. (2021). Determination of critical gel-sol transition point of highly concentrated micellar casein concentrate using multiple waveform rheological technique. *Food Hydrocolloids*, 120.
- Sharma, D., & Sharma, P. (2022). Taguchi design-based synthesis and structural analysis of Cassia galactomannan hydroxypropyl derivative. *Carbohydrate Polymers*, 292, Article 119672.
- Shi, X.-D., Yin, J.-Y., Cui, S. W., Wang, Q., Wang, S.-Y., & Nie, S.-P. (2020). Plant-derived glucomannans: Sources, preparation methods, structural features, and biological properties. *Trends in Food Science & Technology*, 99, 101–116.
- Silva, C., Torres, M. D., Chenlo, F., & Moreira, R. (2017). Rheology of aqueous mixtures of tragacanth and guar gums: Effects of temperature and polymer ratio. *Food Hydrocolloids*, 69, 293–300.
- Silveira, M. L. L., Smiderle, F. R., Moraes, C. P., Borato, D. G., Baggio, C. H., Ruthes, A. C., ... Iacomini, M. (2014). Structural characterization and anti-inflammatory activity of a linear β -D-glucan isolated from *Pleurotus sajor-caju*. *Carbohydrate Polymers*, 113, 588–596.
- Suh, K. Y., Yang, J. M., Khademhosseini, A., Berry, D., Tran, T.-N. T., Park, H., & Langer, R. (2005). Characterization of chemisorbed hyaluronic acid directly immobilized on solid substrates. *Journal of Biomedical Materials Research. Part B, Applied Biomaterials*, 72(2), 292–298.
- Sun, B., Zhang, W., Liu, Y., Xue, M., Qiu, L., & Meng, Z. (2022). A biomass based photonic crystal hydrogel made of *Bletilla striata* polysaccharide. *Biosensors*, 12(10).
- Synytysa, A., Choi, D. J., Pohl, R., Na, Y. S., Capek, P., Lattová, E., ... Park, Y. I. (2015). Structural features and anti-coagulant activity of the Sulphated polysaccharide SPS-CF from a green alga *Capsosiphon fulvescens* (Vol. 17(6), pp. 718–735). New York, N.Y.: Marine biotechnology.
- Tan, L., Liu, S., Li, X., He, J., He, L., Li, Y., ... Guo, J. (2023). The large molecular weight polysaccharide from wild *Cordyceps* and its antitumor activity on H22 tumor-bearing mice. *Molecules (Basel, Switzerland)*, 28(8).
- Tang, Y., Minhui, H., Tang, F., Huang, R., Wang, H., Dingcai, W., & Lan, P. (2022). Easily-injectable shear-thinning hydrogel provides long-lasting submucosal barrier for gastrointestinal endoscopic surgery. *Bioactive Materials*, 15, 44–52.
- Tang, Z., Zhang, M., Gao, L., Bao, Y., Li, P., Wang, M., ... Liu, C. (2023). Optimal extraction of polysaccharides from *Stevia rebaudiana* roots for protection against hydrogen peroxide-induced oxidative damage in RAW264. 7 cells. *Natural product research*, 1–5.
- Tang, Z., Dan, N., & Chen, Y. (2024). Utilizing epoxy *Bletilla striata* polysaccharide collagen sponge for hemostatic care and wound healing. *International Journal of Biological Macromolecules*, 259(Pt 1), Article 128389.
- Tian, W., Huang, Y., Liu, L., Yuanshan, Y., Cao, Y., & Xiao, J. (2024). Tailoring the oral sensation and digestive behavior of konjac glucomannan-gelatin binary hydrogel based bigel: Effects of composition and ratio. *International Journal of Biological Macromolecules*, 256, Article 127963.
- Torres, M. D., Hallmark, B., & Wilson, D. I. (2014). Effect of concentration on shear and extensional rheology of guar gum solutions. *Food Hydrocolloids*, 40, 85–95.
- Wang, B., Zhang, H., Chen, L., Mi, Z., Yan, X., Zhao, G., ... Niu, J. (2020). Extraction, purification, and determination of the gastroprotective activity of glucomannan from *Bletilla striata*. *Carbohydrate Polymers*, 246, Article 116620.
- Wang, L., Yujia, W., Li, J., Qiao, H., & Di, L. (2018). Rheological and mucoadhesive properties of polysaccharide from *Bletilla striata* with potential use in pharmaceuticals as bio-adhesive excipient. *International Journal of Biological Macromolecules*, 120(Pt A), 529–536.
- Wang, N., Zhang, K., Chen, Y., Jie, H., Jiang, Y., Wang, X., & Ban, Q. (2023). Tuning whey protein isolate/hyaluronic acid emulsion gel structure to enhance quercetin bioaccessibility and in vitro digestive characteristics. *Food Chemistry*, 429, Article 136910.
- Wang, Q., Haiming, H., Xiong, L., Jin, X., Zhu, T., Sun, X., ... Liu, H. (2024). Structural characterization and prebiotic activity of *Bletilla striata* polysaccharide prepared by one-step fermentation with *Bacillus Licheniformis* BJ2022. *International Journal of Biological Macromolecules*, 258(Pt 1), Article 128822.
- Wang, Y., Liu, D., Chen, S., Wang, Y., Jiang, H., & Yin, H. (2014). A new glucomannan from *Bletilla striata*: Structural and anti-fibrosis effects. *Fitoterapia*, 92, 72–78.
- Wang, Y., Han, S., Li, R., Cui, B., Ma, X., Qi, X., ... Li, S. (2019). Structural characterization and immunological activity of polysaccharides from the tuber of *Bletilla striata*. *International Journal of Biological Macromolecules*, 122, 628–635.
- Wei, Y., Guo, Y., Li, R., Ma, A., & Zhang, H. (2021). Rheological characterization of polysaccharide thickeners oriented for dysphagia management: Carboxymethylated curdlan, konjac glucomannan and their mixtures compared to xanthan gum. *Food Hydrocolloids*, 110.
- Xiang, J., Wang, Y., Yang, L., Zhang, X., Hong, Y., & Shen, L. (2022). A novel hydrogel based on *Bletilla striata* polysaccharide for rapid hemostasis: Synthesis, characterization and evaluation. *International Journal of Biological Macromolecules*, 196.
- Yang, L., Han, Z., Chen, C., Li, Z., Shaopeng, Y., Yan, Q., & Zeng, R. (2020). Novel probiotic-bound oxidized *Bletilla striata* polysaccharide-chitosan composite hydrogel. *Materials Science & Engineering. C, Materials for Biological Applications*, 117, Article 111265.
- Ye, G., Li, J., Zhang, J., Liu, H., Ye, Q., & Wang, Z. (2021). Structural characterization and antitumor activity of a polysaccharide from *Dendrobium wardianum*. *Carbohydrate Polymers*, 269, Article 118253.
- Yuan, L., Zhong, Z.-C., & Liu, Y. (2020). Structural characterisation and immunomodulatory activity of a neutral polysaccharide from *Sambucus adnata* wall. *International Journal of Biological Macromolecules*, 154, 1400–1407.
- Zhang, H., Li, C., Lai, P. F. H., Chen, J., Xie, F., Xia, Y., & Ai, L. (2021). Fractionation, chemical characterization and immunostimulatory activity of β -glucan and galactoglucan from *Russula vinosa* Lindblad. *Carbohydrate Polymers*, 256, Article 117559.
- Zhang, M., Qin, H., An, R., Zhang, W., Liu, J., Yu, Q., ... Huang, X. (2022). Isolation, purification, structural characterization and antitumor activities of a polysaccharide from *Lilium davidii* var. *unicolor* cotton. *Journal of Molecular Structure*, 1261, Article 132941.
- Zhang, Y., Wang, H., Guo, Q., Wang, J., & Cui, S. W. (2020). Structural characterization and conformational properties of a polysaccharide isolated from *Dendrobium nobile* Lindl. *Food Hydrocolloids*, 98, Article 104904.
- Zhao, H., Qiang, X., Yan, T., Zhang, H., & Yang, Y. (2023). Effect of *Bletilla striata* polysaccharide on the pasting, rheological and adhesive properties of wheat starch. *Polymers*, 15(24).
- Zhao, Y., Wang, D., Jingxin, X., Dongkun, T., Zhuang, W., & Tian, Y. (2024). Effect of polysaccharide concentration on heat-induced *Tremella fuciformis* polysaccharide-soy protein isolation gels: Gel properties and interactions. *International Journal of Biological Macromolecules*, 262, Article 129782.
- Zhou, S., Huang, G., & Huang, H. (2022). Extraction, derivatization and antioxidant activities of onion polysaccharide. *Food Chemistry*, 388, Article 133000.